

IN RE: NYAYA-AI

The Case for Nyaya-AI.



Every legal-AI demo can draft an argument. This memorial argues why Nyaya-AI is the one built so it structurally cannot draft a citation that doesn't exist.

STATEMENT OF FACTS

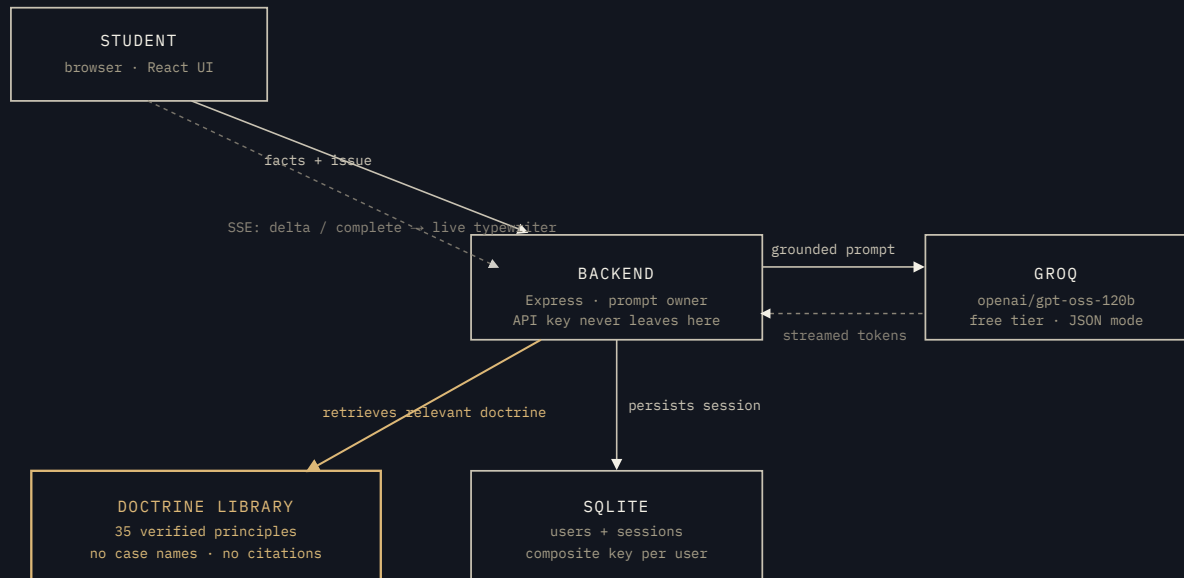
The gap in moot-court preparation.

¶ 1 Legal-research platforms are priced and built for practicing lawyers, not first-year students learning to structure an argument. The obvious shortcut — asking a general chatbot — drafts confidently and cites fabricated case law just as confidently.

¶ 2 Nyaya-AI is a practice sandbox: paste a fact pattern, receive a structured IRAC argument, an opposing counsel's rebuttal, and a plain-language translation — engineered from the system prompt up so the model is architecturally restricted to doctrine it can name, never a case it can invent.

EXHIBIT A

How a request actually moves through the system.



The retrieval hop (brass) is the whole argument: the model is grounded in a hand-verified doctrine library before it ever writes a word – never in a database of case law it could misquote.

Where this stops looking like a hackathon prompt wrapper.

FINDING I Retrieval without importing the risk

Most “RAG for legal AI” builds mean a vector database of scraped case law — which just relocates the hallucination risk from invented to unverifiable. Nyaya-AI retrieves only from a small, hand-written library of general doctrine and grounds the model in that before generation starts. Verified live: ask about a mislabeled price, and the model’s own output names “*Unilateral Mistake (Snapping Up)*” — retrieved, not invented.

THE OBVIOUS BUILD

Pinecone + scraped judgments, cited with false confidence.

THIS BUILD

35 verified doctrines, zero case law, zero fabrication surface.

FINDING II You can watch it reason

POST /api/generate speaks Server-Sent Events. Tokens arrive live from Groq, through the backend, into the browser, typed out in real time behind a blinking cursor — not a spinner standing in for three seconds of silence.

THE OBVIOUS BUILD

Loading spinner, then a wall of text appears.

THIS BUILD

Token-level SSE streaming, rendered live as it’s written.

FINDING III The guest login isn't a trick

"1-Click Instant Access" issues a real, backend-signed JWT for an anonymous session — the same bcrypt/JWT pipeline, the same database-backed history as a registered account. There is no separate fake demo mode quietly standing in for the real product.

THE OBVIOUS BUILD

setTimeout pretending to be a login.

THIS BUILD

A real, server-verified guest session, indistinguishable in storage.

FINDING IV A real database, not a folder of JSON

Session history lives in SQLite via Node's own built-in driver — zero extra dependencies — with a composite primary key that makes one user's session id structurally unable to collide with another's. No row cap, no flat files.

THE OBVIOUS BUILD

localStorage, capped, unstructured.

THIS BUILD

Indexed SQL, per-user isolation enforced by the schema.

FINDING V 45 tests, not a demo that only works once

28 backend tests and 17 frontend tests cover auth, per-user isolation, RAG retrieval, streaming validation, and export formatting — run on every change, not written after the fact for the eval.

THE OBVIOUS BUILD

Tested by demoing it once, live, and hoping.

THIS BUILD

45 automated tests, green before every submission.

EXHIBIT C

The record.

AI EVALUATION SCORE100 / 100 · A+

AUTOMATED TESTS PASSING4528 backend · 17 frontend

COST TO RUN\$0free-tier Groq only, no paid API

LEGAL SUBJECTS COVERED7Contract, Tort, Criminal, Constitutional ...

RETRIEVAL KNOWLEDGE BASE35hand-verified general doctrines

A ninety-second script for the judges' table.

- 1 Open already loaded on the sample contract-mistake scenario — don't type live, judges have seen that stall.
- 2 Press **Generate** and stop talking. Let the argument type itself onto the screen in real time.
- 3 Point at one retrieved principle name and say it plainly: **“that name isn't the model guessing — it was retrieved from a doctrine library we hand-verified before generation even started.”**
- 4 Chain **Counterargument** then **Score My Argument** in one breath — show the pipeline, not one lucky output.
- 5 *Close with the one sentence that survives Q&A: “We didn't just avoid hallucination — we built the one architecture where hallucinating a citation was never on the table.”*

Prepared as submission collateral for the judges' table.

Not part of the graded repository — filed separately, at no risk to the record above.

§ EXHIBIT FILED · CASE RESTS